

Brokers Who Do Not Bridge: When Assessment, Not Access, Shapes Broker Advantage

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Abstract

Structural theories describe economic brokerage as an opportune but fragile position bridging gaps between otherwise disconnected actors. In that view, brokering can undermine trust or eliminate the gaps that generate its value, and brokers need supporting institutions or deeper interpersonal ties to stabilize their role. Yet many corporate actors in positions of brokerage, from Visa to Amazon, have risen in market power, profit, and prominence even as networks transformed around them. When can brokering strengthen rather than undermine a broker's advantage?

I reconceptualize brokerage as outsourced relational work, in which a broker who assumes no inventory risk constructs viable matches between parties who cannot easily find or evaluate one another. Repeated matchmaking generates an informational byproduct that the broker can accumulate: knowledge of what makes pairings successful across the market, unavailable to clients who observe only their own dealings. An agent-based model of a matching market identifies conditions under which this informational advantage emerges and persists. Brokerage can provide assessment rather than access: clients outsource matching even when they can reach counterparties themselves, because the broker's predictions are valuable. Counterintuitively, the broker can become highly central in regimes where it engages in little bridging behavior: broker centrality is a product, not a primary source, of matchmaking quality.

While the model explores the dynamics of the broker's informational advantage, I theorize how this advantage can then enable the broker to become a principal selling the resource it once intermediated or monetizing data and analytics. This is *transient brokerage*, a process that highlights a broker's growing power rather than its fragility.

I Introduction

Intermediation in economic relations—the intervention of a third party to mediate or facilitate exchange between others—can take many forms. Brokerage is one of them. In markets, brokers are intermediaries that do not assume inventory risk: they act as matchmakers, connecting others.

Brokerage is pervasive in a wide variety of markets and remarkably persistent across time and space. Brokers find themselves in an advantageous but delicate position. They need to stand in-between disconnected others to have the opportunity to broker, but through their work, they densify the networks around them, potentially eroding their own advantage. Unlike dealers and wholesalers, brokers do not maintain an inventory that they need to manage and can liquidate in case of difficulty. Unlike market makers, they keep offers and price information private (Rust and Hall 2003). They operate at the intersection of social groups (Burt 1992, 2004) and rely on secret information. As a result, they often lose the trust and legitimacy needed to continue their activities. Prominent theories of brokerage have described it as an inherently fragile role that often gets “stabilized” to avoid collapsing, through the social isolation of brokers, their organizational grafting, or their capture by one side (Stovel, Golub, and Milgrom 2011).

If that is the case, then how have so many corporate actors in positions of brokerage, from Visa to Amazon, managed to rise in market power, profit, and prominence even as networks transformed around them?

Recent work by Ronald S. Burt, building on the tradition of structural-hole brokerage theory, has shown how many bridging ties are not weak or fragile, but rather strong high-trust ties forged through interpersonal history, which provide a robust and enduring advantage to the broker who relies on them (Burt 2026; Burt and Oppen 2026). However, this is not the full story of broker strength.

Some brokers leverage their role to become principals rather than remain only intermediaries. They stabilize their advantage while their brokering work becomes secondary or even stops entirely. I call this *transient brokerage*, a phenomenon that current theories of brokerage do not satisfactorily predict or explain. Brokering sometimes declines or disappears because of the broker's power, not its fragility. This article explains under what conditions this happens.

This article redefines brokerage as *outsourced relational work*: the broker constructs viable matches between parties who cannot easily find or evaluate each other. This relational work generates an informational byproduct that the broker can leverage. While structural position can provide access to feed learning, each successful match between previously unconnected parties can erode the broker's structural advantage by densifying the network (Burt 2000, p. 356), but it also helps the broker accumulate knowledge of what makes pairings successful. In that way, the broker converts structural capital into informational capital through the act of brokering. This is particularly true for corporate brokers, who operate at scale and benefit from the legal privileges of the corporate form.

This article develops a theory of brokerage as outsourced relational work and explains how the resulting informational advantage can enable a transition from intermediary to principal. The article addresses three questions about brokerage:

- (i) What value does a broker provide?
- (ii) How does brokering alter a broker's informational and structural position?
- (iii) Under what conditions can the broker convert its advantage to become a principal?

I develop an agent-based model (ABM) of brokerage in a general matching market to formalize the basis for the theory and illustrate how brokers shape and sustain structural and informational advantages through repeated matchmaking. The simulations show how several simple elements interact to produce non-obvious dynamics, complicating the expectation that brokers densify networks around them and erode their own position, and producing propositions for future empirical research. The ABM results suggest that brokerage can provide assessment rather than access, and that clients may continue outsourcing even when they can reach counterparties themselves. An experiment separating the two services shows that assessment contributes more value to principals than access.

The broker ranks potential matches better than its clients. This assessment advantage comes from the information generated through its relational work. By pooling observations across clients, the broker learns how both parties' characteristics combine. These assessments are more valuable when the output of a realized match depends on complementarity, or how well two parties fit together, rather than the general quality of each party only. By providing valuable assessments, the broker can sustain demand for its relational work, and that sustained use can increase its structural prominence. Bridging position is distinct from bridging behavior: a broker can be highly central while doing little or no bridging. The ABM results recast betweenness centrality not only as a source of brokerage opportunity, but also as a product of the kind and quality of a broker's relational work.

I build on these results to theorize how brokers can turn data obtained through their activity into a business more profitable than intermediation itself. Depending on the market in which the broker operates, the nature of the matching problem, and the relations between the actors involved, the

broker can leverage its information advantage to convert relational work done on behalf of others into personal economic advantage. When successful, **the broker becomes a principal selling the resource it once intermediated or monetizing data and analytics**. This explains the rise in prominence of some firms acting as brokers.

The rest of this article proceeds as follows. Section II reviews structural and processual theories of brokerage and their blind spots. Section III reconceptualizes brokerage as outsourced relational work and traces how this work is transformed when performed at scale by corporate actors. Section IV presents the ABM and its results illustrating how a broker produces and sustains its advantage through this work. Section V builds on this foundation to theorize one possible trajectory that can follow from the broker advantage, transient brokerage, under which a broker can become a principal. Section VI discusses scope conditions and implications for the political economy of corporate intermediation, and Section VII concludes.

II Theories of Brokerage and their Blind Spots

In markets where information is scarce or complex, participants are few, transactions are intricate, or institutions are poorly developed (Stovel, Golub, and Milgrom 2011), participants tend to seek the help of third parties to facilitate their transactions. Economic brokerage thus tends to emerge when counterparties are hard to find, resources hard to value, or transactions hard to structure, making intermediation work valuable. Real estate, mortgage, stock, and commodity markets are common examples of markets with many active brokers; so are markets for rare items and collectibles, over-the-counter financial markets, and many others.

Brokerage may also be helpful in circumstances where tight-knit, closed groups monopolize resources or information, and outsiders cannot easily access them (Stovel and Shaw 2012, p. 140). Brokerage can maintain or reinforce group boundaries, or it can blur or collapse them (Stovel and Shaw 2012), with consequences for social segregation, social integration, and relations between groups or communities (Galster and Godfrey 2005).

Past research has proposed two conflicting sets of definitions of brokerage (see Erikson 2013, for a review): formalist or structuralist definitions (Burt 1992; Gould and Fernandez 1989; Granovetter 1973; Marsden 1982), and broader processual definitions (Obstfeld, Borgatti, and Davis 2014; Rossman 2014; Spiro, Acton, and Butts 2013; Stovel and Shaw 2012).

A Formalist and Structuralist Definitions

Early brokerage research examined the structure of networks and the positions of actors to identify brokerage forms, categorize them, and determine their consequences (Burt 1992, 2004; Gould and Fernandez 1989; Granovetter 1973; Marsden 1982; Simmel 1964, 1971). Brokerage opportunities arise when there are gaps in a social network, such that some nodes are close (in degrees of separation) yet disconnected (Burt 1992, 2004; Marsden 1982). Brokerage is synonymous with open triads or structural holes: a structure in which a third actor bridges two otherwise unconnected nodes. The third actor mediates flows of resources and information and facilitates transactions between others who lack access to one another (Fernandez and Gould 1994, p. 1457; Marsden 1982, p. 202). Researchers in this tradition generally set aside interactional content to achieve greater generality (Erikson 2013). For instance, Burt (1992, p. 36) was explicit in treating “motivation and opportunity as one and the same.”

Burt (1992) operationalized the structural-hole concept, proposing the measures of network constraint and effective size to quantify how well positioned an actor is for brokerage, which allowed the theory to be tested across a wide range of settings. For instance, in a follow-up study, Burt (2004) investigated

the network position of 673 managers in a large electronics company. He found that those who spanned structural holes generated ideas that senior management judged more valuable and were rewarded with better performance evaluations, promotions, and compensation. He suggested that brokers appear more creative because they can take insights that are commonplace in one group and introduce them to other people to whom they appear novel and original (Burt 2004, p. 388).

As such, brokers derive an advantage from their ability to exploit pre-existing information in the network. Because of their position, brokers get privileged access to diverse ideas from across the network. They can see opportunities that others do not, and they can strategically use their position by providing or withholding information (Burt 1992) or using it to sustain their own creative endeavors. Exposed to varied perspectives, they are well-positioned to build a richer understanding, synthesize across domains or outlooks, and develop a unique vision (Burt 2004).

Defining brokerage based on structural holes, however, excludes situations that clearly appear as brokerage to a lay observer. For instance, Alice wants to sell her house and asks a real estate broker to list it. Soon after the real estate broker puts up the ‘for sale’ sign, her neighbor Bob, whom she knows personally, contacts the broker to purchase the house. A broker intermediates the transaction, yet no structural hole existed between buyer and seller.

B Processual Definitions

More recently, some researchers have recast brokering as a set of relational processes (Obstfeld, Borgatti, and Davis 2014; Rossman 2014; Spiro, Acton, and Butts 2013; Stovel and Shaw 2012). Stovel and Shaw suggested that brokerage was “the process of connecting actors in systems of social, economic, or political relations in order to facilitate access to valued resources” (Stovel and Shaw 2012, p. 141; see also Stovel, Golub, and Milgrom 2011). Brokers exhibit two crucial features: they are positioned as bridges across gaps, and they help transfer resources or knowledge across those gaps (Stovel and Shaw 2012, p. 141). This definition remains constrained by social structure but requires brokerage behavior. Hence, there can be structural holes without brokerage—some bridging actors do not transmit anything across holes—but no brokerage without structural holes. In this way, it runs into the same conceptual problem as purely structural definitions.

Obstfeld, Borgatti, and Davis (2014), building on Marsden (1982), proposed a purely processual definition: brokerage is a process by which an actor “alters interaction between two or more parties in a wide variety of triadic structures” (Obstfeld, Borgatti, and Davis 2014, p. 136), meaning that he “influences, manages, or facilitates” the interaction (Obstfeld, Borgatti, and Davis 2014, p. 141). This eliminates any structural requirement beyond the need for three or more actors. It is no longer necessary for the alters to lack access to or trust in each other. There can be structural holes without brokerage, and there can also be brokerage without structural holes: brokerage behavior can, and often does, occur in dense networks.

Previous work on brokerage, particularly in the processual tradition, has also identified categories of brokers depending on the nature of their activities. Middleman, transfer, or conduit brokers facilitate economic transactions without necessarily altering the structure of a local network, like stockbrokers (Stovel and Shaw 2012, p. 146; Spiro, Acton, and Butts 2013; Obstfeld, Borgatti, and Davis 2014, p. 141). Catalyst, moderator, or matchmaking brokers focus on creating lasting ties, like headhunters and party hosts (Stovel and Shaw 2012, p. 146; Obstfeld, Borgatti, and Davis 2014; Spiro, Acton, and Butts 2013). Finally, *coordinators*, such as middle managers or disaster response teams, work to transform existing ties by reconfiguring local networks of interdependence (Spiro, Acton, and Butts 2013, pp. 131–132).

While definitions based on structural holes were very restrictive, Obstfeld and co-authors’ processual definition is sweeping. For instance, they suggested that, under their definition, a manufacturing firm acts as a broker when it buys raw materials from an alter, transforms them into a product,

and sells the finished product to another alter (Obstfeld, Borgatti, and Davis 2014, p. 145). Most triadic or polyadic relations, in this account, imply brokerage. Indeed, many descriptions of brokers in the literature are so broad as to invite the charge that a concept that explains everything explains nothing. The most varied social facts have been described as acts of brokerage: mediating conflict (Obstfeld, Borgatti, and Davis 2014), reselling stolen property (Steffensmeier 1986), advocating on behalf of someone, making sense of the world for someone, and helping connect epistemic communities (Stovel and Shaw 2012, p. 140), to name only a few.

C Shortcomings of Existing Frameworks

Work that relies chiefly on the structural holes idea generally makes clear that transferring knowledge and ideas is a key element of brokering and of the broker advantage. Yet the information under consideration is generally preexisting and ambient, not generated by brokering itself (Burt 2004). Moreover, the structural account leaves the actual work of brokering largely implicit. Structural position indicates places where brokerage is possible, but not whether anyone is doing the work of brokering—work for which structural holes are neither necessary nor sufficient.

Meanwhile, broad processual definitions blur the lines between vicarious and personal relational work. They include under the same umbrella cases where the intermediary deals with others on its own account, and cases where the intermediary works on behalf of others. As such, this perspective does not examine the informational byproduct that results from repeated vicarious work. Information is often simply listed among the resources that can be transferred through a broker, along with material or financial resources (see, e.g. Obstfeld, Borgatti, and Davis 2014, p. 141; Stovel and Shaw 2012, p. 141).

Neither perspective offers a cohesive theory of the role of information in brokerage, even though information shapes every stage of the brokering process. Before brokerage, the distribution of information shapes the opportunity or need for it: brokerage arises partly out of informational inequality, and the broker exploits what it knows that one or both sides do not. During brokerage, the transmission of information is part of the brokerage act itself, as the broker shares enough to make a match viable while withholding enough to preserve its advantage. After brokerage, the distribution of information has been altered, and the work has generated a byproduct the broker retains, which may shape the broker’s opportunities. The circulation of information is at once a determinant of brokerage, a component of the work of brokering, and a consequence of brokerage.

This article reconceptualizes brokering as outsourced relational work and builds on this theoretical move to explain how brokerage sometimes weakens or disappears because a powerful broker captures resources or encloses data to become a principal.

III Outsourced Relational Work and the Broker Advantage

Social actors engage in *relational work* when they try to interpret and match social ties, economic transactions, and media of exchange, to create viable matches that will shape the outcome of their interaction (Zelizer 2012, p. 151, 2005; see also Bandelj 2020, p. 255). These matches and their shared meanings form distinct *relational packages*. For instance, in a marketplace, actors trading directly with one another draw on cultural knowledge to reach a shared understanding of their situation and the ‘rules of the game,’ and to agree on terms, including price (Block 2012, p. 137).

For the broker’s clients, brokerage is *outsourced relational work*. For the broker, brokerage is *vicarious relational work*. In economic markets, this outsourced relational work is usually performed at scale by a *corporation*. This is more than a question of definitions: it shapes the broker’s opportunities, its strategic advantage vis-à-vis its clients, and its trajectory.

A From Personal to *Outsourced and Vicarious* Relational Work

In brokered transactions, brokers take on the work of constructing a shared reality for limited purposes. The core of what brokers do is relational matching (see Zelizer 2012). Brokers are characterized by the relational work they do *on behalf of others*.

For the broker, this vicarious work requires another layer of relational work to negotiate brokerage relations with its clients. Brokerage corresponds with a distinct relational package. Not all intermediation is brokerage. In the relational package of economic brokerage, the broker performs a matching service for others, usually for a fee, commission, or retainer. Brokers bundle various services and amenities as part of that package: they take on the efforts of search and negotiation, they provide access to otherwise inaccessible counterparties, they share with their clients their expertise and the benefits of the private information they hold about quality, pricing, and availability, and they document and facilitate transactions, which often require logistical, bureaucratic, or regulatory work.

The middleman, catalyst, and coordination brokerage described in Section II are not exclusive categories of brokers, but rather, components of the work of brokering, which involves a bit of each in varying proportions. An important part of brokers' relational matching is to facilitate interaction (middleman brokerage), but this work also directly impacts interpersonal relations and local networks, often in an enduring way, by creating new and lasting ties (catalyst brokerage) or by reorganizing local networks (coordination brokerage).

Brokerage inherits the dual character of relational work, which can both foster and reduce informational asymmetry when constructing shared understandings while maintaining strategic opacities (Block 2012, p. 138). In performing relational work on behalf of others, the broker simultaneously clarifies and obscures: it transmits enough information to make a transaction legible and a match viable and conceals enough to justify its fee and preserve its advantage. The same opacity can do social as well as economic work. Rossman (2014) shows how brokerage functions as obfuscatory relational work that manages the taboos around disreputable exchange. Outsourcing relational work provides cover to parties in morally fraught matches. What the broker withholds is also the raw material of its later advantage: information that it withholds, accumulates, and, under the right conditions, eventually monetizes.

Vicarious work relates to entities or resources that are *not* under the broker's personal ownership and control, and the shared understanding of the parties is that the broker is *not a principal* in the brokered transaction (see, e.g. Stovel, Golub, and Milgrom 2011, p. 21326). Brokers generally have limited personal involvement in brokered interactions beyond their role as facilitators. The broker is a supporting actor, not a main character. Intermediaries who have full control over traded resources occupy a very different position in terms of risk, power, and vulnerability.

Another way to put it is that in economic markets, brokers are intermediaries that *do not assume inventory risk*. That is, if Bob helps Alice and Charlie conclude a sale, Bob is a broker. If Bob buys from Alice and eventually leases to Charlie, he is not. For instance, a real estate broker using her market knowledge to buy houses in up-and-coming neighborhoods to rent to families no longer acts as a broker. Instead, the relational work she performs is personal: matching herself to sellers and renters.

B From Individual to *Corporate* Relational Work

The definition above applies naturally to describe individual brokers working with individual clients, such as a real estate agent, an insurance broker representative, or a matchmaker. Much of the work of brokerage, however, occurs at the corporate, organizational, or institutional level, where brokers are firms, and clients may also be.

To understand what happens when relational work is outsourced to corporations, we must first discuss how corporations perform relational work in the first place, for themselves as organizational social actors. Relational work scholarship initially focused on the intimate sphere and interpersonal relationships (see Bandelj 2020, for a review). But relational work also takes place in the economic sphere and at the organizational level. The relational work tradition has acknowledged the organizational question without fully resolving it. Bandelj (2012) noted that relational work between representatives of organizations solidifies into macrolevel structures, while Zelizer (2012, pp. 156–157) catalogued instances of relational work by firms in the field of labor relations. Bandelj’s (2020) review of “scaling up” relational work surveyed studies operating at the organizational level, but they mainly focused on what people do with and to each other within organizations, not on the corporate form itself as a vehicle for relational work.

Firms must conduct relational work to manage contingent, fleeting ties and longer-term relationships with workers, consumers, suppliers, lenders, investors, bureaucrats, and so on. The properties of the corporate form matter to this work and make it distinct. The firm is not simply a site where interpersonal relational work occurs; nor is it a singular, scaled-up actor that performs relational work as an individual does.

Firms are groups of people in designated roles, following norms and practices, supported by procedures, interfaces, and infrastructures, and doing the everyday work of managing the firm’s relationship with its environment “at the coalface” (Gray and Silbey 2014; Huising and Silbey 2018). The mining metaphor directs attention to the frontline of the organization, where people enact its dealings with the outside world and translate organizational imperatives into routine practice. Relational work, like everything the corporation does in the world, is enacted through the people, procedures, infrastructure, and algorithms working at the coalface. Because of its collective and distributed nature, it is never visible in its entirety to any of the people involved.

Most firms are also legally organized as corporations. The corporation is a political entity with emergent capacities for coordination, information processing, and strategic action that exceed the sum of its parts (Ciepley 2013; Farrell and Newman 2023). There is a fundamental asymmetry between corporate actors and natural persons (Coleman 1982, 1990). The corporation aggregates information across thousands or millions of interactions, while each natural person sees only their own. The corporation can persist indefinitely, while individual relationships are transient. The corporation coordinates strategically across its full portfolio of relationships, while each counterparty acts in isolation. As a result, when a corporate actor interacts with natural persons, the corporate actor has structural and informational advantages.

These asymmetries are not accidental byproducts of size: they are legally constituted features of the corporate form. The corporation is one of the core modules of the legal code of capital (Pistor 2019). Corporate law endows the firm with durability that outlasts any individual relationship, priority that ranks its claims above those of other parties, and universality that makes those claims enforceable against the world, so that it can become a legal technology for concentrating and protecting accumulated advantages and leveraging them for productive purposes. These are the conditions under which corporate relational work takes place: it is inherently asymmetric, because of the different scales and material and informational resources of corporate and natural persons.

C Outsourcing Relational Work to Corporations: The Corporate Brokers

When a client outsources relational work to a corporate broker, the corporate broker performs that work through people and practices at the coalface, each working in some small way to define relational packages, perform appropriate matches, and shape information flows. The corporate form reorganizes vicarious relational work, distributes it across roles, and embeds it in routines and procedures. To

orchestrate this, corporate brokers develop an organizational apparatus comprising internal and client-facing roles, information architectures and databases, contractual templates, and interfaces.

Saying a brokerage firm does relational work on behalf of its clients means that in sales meetings, in contract negotiations, in the design of user interfaces, in the defaults embedded in terms of service, and so on, people relying on organizational procedures are working to define the features of its clients' relationships, what transactions are appropriate to them, and what media may govern the exchanges. Just like for its own relational work, the corporation has a capacity for coordination, information aggregation, and temporal persistence that makes the vicarious relational work of corporate brokerage qualitatively different from what any individual could accomplish alone.

The corporate broker's relation with its clients is structurally asymmetric. With human clients, corporate and natural persons bring fundamentally different legal, informational, and temporal architectures to the relationship. Even with corporate clients, however, the asymmetry remains: the broker coordinates vicarious relational work strategically and benefits from cross-market information, while the client encounters it piecemeal. The matching of relations, transactions, and media moves from a pure negotiation between principals with comparable standing, who learn from each transaction, toward an administered process in which the corporate broker sets the terms, explicitly or behind the scenes, and captures the information that each pairing generates.

The data infrastructure of the corporate broker includes data taxonomies, client categories, risk models, algorithmic sorting procedures, and so on. This infrastructure does not merely facilitate transactions but actively constitutes relational packages by determining what transactions are possible, and what categories they should fall into. Classification systems and information infrastructures, like those of the corporate broker, are not neutral descriptions of the world; they actively organize social relations (Bowker and Star 1999; Fourcade and Healy 2024). The corporate broker's information infrastructure, much like a corporate rating agency's credit-scoring system or a platform's matching algorithm, sorts counterparties into categories the broker defines and uses those categories to allocate opportunities according to the broker's objectives. The impact of the broker's information infrastructure is magnified because the corporate broker works at scale, while individuals cannot usually design categories that simultaneously structure millions of relationships.

D The Broker Advantage

Why do people and firms outsource their relational work to brokers? To have the opportunity to broker, a participant in a matching market must hold either a *structural* or an *informational* advantage. A structural advantage arises from network position and can be used to provide *access*: the broker bridges by bringing together parties that were not previously connected. An informational advantage supports *assessment*: the broker evaluates prospective matches better than its clients can. The broker's service is therefore valuable when it helps clients reach counterparties they could not otherwise find or better evaluate counterparties already available to them.

Assessment involves two components: evaluating the general quality of the goods or counterparties involved and evaluating the quality of the specific pairing or relational package being considered. These components also correspond to two possible channels for the broker's information advantage: *attributional*, about evaluating how parties' attributes translate into individual quality, and *relational*, about evaluating which counterparties best complement each other. Both are self-reinforcing, as the broker accumulates the knowledge required for assessment through the relational work it performs, and that learning makes it better at the work. The broker observes outcomes from many pairings across many principals, while each principal observes only its own. This gives the broker both a cross-market view and higher data volume.

The broker's information work can be informal and internal, as with a real estate broker who knows the local market, has an intuition for the right asking price, and holds social ties to likely buyers. It

can also be formal and bureaucratized, as with a corporate mortgage broker that follows tables and formulas to determine which interest rate may produce successful matches between borrowers and lenders. The latter records and retains what the former holds in personal notes or experience.

Important models in the economics literature have focused on the attributional channel and characterized the broker’s role primarily as quality certification (Li 1998) or expert screening (Bethune, Sultanum, and Trachter 2022): the broker identifies which goods or counterparties are high quality. In that view, the broker is an appraiser whose cross-market experience helps it assess the general quality of counterparties more accurately than individual principals can. In contrast, the relational-work view of brokerage suggests that the broker’s value lies also, and often primarily, in understanding complementarities between counterparties and shaping relational packages accordingly. The broker is a relational worker and a matchmaker whose advantage comes from the ability to make pairings and to know which pairings will likely succeed, an ability that improves with repeated matchmaking.

The model that follows illustrates the basis for the theory: a repeat-player broker observes outcomes across clients and uses the resulting information to organize subsequent matches. The model portrays how the broker’s work, outsourcing rates, and network topology can evolve together.

IV How Brokers Accumulate Informational and Structural Advantage: A Formal Illustration

A Model and Methods

I use an agent-based model (ABM) to formalize the theory. The model is an illustration, not a test of the theory; it serves as a bridge from theory to testable propositions. Complete pseudocode and specifications appear in Appendices A and B. The public git repository (<https://github.com/m-laprise/brokerage-abm>) contains the Julia code and scripts needed to reproduce the simulations, statistics, figures, and manuscript text.

The model isolates the informational consequences of outsourced relational work. Principals learn only from their own matches, whereas the broker pools information from matches it mediates across principals. Learning to evaluate general quality corresponds with attributional assessment, while learning to evaluate complementarity corresponds with relational assessment. The broker represents the information-aggregating function of corporate brokerage, not its internal organization, and the model does not simulate the later transition to resource capture or data enclosure.

Market and match output. Principals are the market participants seeking partners. Many principals and one broker occupy an undirected network. Each principal has a fixed vector of observable characteristics, which defines its type. At baseline, types are drawn around a smooth curve in an eight-dimensional space of observable characteristics (Supplementary Figure S1). For any potential pair, expected match output is determined by the two principals’ types and by fixed features of the matching environment drawn at the start of each simulation run. A match’s output combines two components: general quality, which depends on each principal separately, and complementarity, which depends on how their types interact (Appendix B §2). Under pure general quality, a higher-quality counterparty raises expected output for every principal. With complementarity, expected output also depends on which two principals are paired (Supplementary Figure S2).

Before they are combined, the components are centered and scaled to have the same variance. The parameter ρ sets their relative contributions to variation in expected match output: general quality contributes a fraction ρ , and complementarity contributes $1 - \rho$. Thus, $\rho = 0$ represents pure complementarity and $\rho = 1$ represents pure general quality. The parameter δ controls the difficulty of learning complementarity by varying how sharply complementarity changes across an arbitrary boundary in type-pair space: it multiplies complementarity by $1 + \delta$ on one side and $1 - \delta$ on the

other. Larger values make otherwise similar pairs on opposite sides of the boundary differ more in expected output. A common normalization holds the average within-market variance of expected match output constant as ρ and δ vary. Because δ affects only complementarity, it has no effect when $\rho = 1$. Describing variation in expected output across pairs requires more dimensions under complementarity than under pure general quality, particularly when difficulty is high (Supplementary Figure S3).

When a pair matches, its realized output equals its expected output plus an independent, mean-zero random shock. This shock represents match-specific variation that neither the principals nor the broker can anticipate.

Search, matching, and learning. Each period, principals seek pairwise matches either through self-search or through the broker (Appendix A §§5–7; Appendix B §§6–7). They choose one channel using satisfaction with past realized output net of search costs or broker fees. Self-search considers existing neighbors and a small random sample of other principals. The broker considers principals in its standing roster (a persistent list of contacts), and those currently using its services.

Principals and the broker accumulate observations and learn in parallel. After every completed match, each principal records its partner’s characteristics and the realized output, whether the match was arranged through self-search or through the broker. The broker also records both parties’ characteristics and realized output from every match it mediates, pooling observations across clients. It does not observe matches arranged entirely through self-search.

Each principal uses its own observations to predict the output of matching with a candidate, using that candidate’s characteristics. The broker uses its pooled observations to predict output from both parties’ characteristics. In the main variant, each principal has its own neural network, and the broker has a separate neural network. In another variant, these predictions use Ridge regression.

Estimated output determines how candidates are ranked. For previously matched neighbors, self-search instead uses the pair’s average past realized output. Tied estimates are ordered uniformly at random. Principals make offers to the highest-ranked candidates whose estimated output exceeds the reservation threshold, the minimum required for an offer. All offers enter a common market. A completed match creates a direct tie that persists until either principal exits. At the end of each period, every principal exits independently with probability η and is replaced by a new entrant.

Measurement. The outsourcing rate is the share of requested match positions routed through the broker. A brokered placement is classified as access when it creates a direct tie between previously unconnected principals and as assessment when it reuses an existing tie. The access fraction is the share of brokered placements classified as access. The output gap is the difference in mean realized output between brokered matches and self-search matches. Net output to principals is realized output net of search costs and broker fees, averaged over all requested match positions, including those that remain unfilled. Broker betweenness centrality measures the share of shortest paths between principals that pass through the broker (Brandes 2001; Freeman 1977). Supplementary Figures S5–S7 reproduce the structural analyses using Burt’s constraint and effective size instead of betweenness centrality.

Prediction quality is evaluated each period using randomly sampled potential partners. These evaluations do not update either model. Predictions are compared with the expected output, without random shock. Rank correlation is Spearman’s correlation between the candidate ordering produced by the predicted output and the ordering produced by the expected output. The ranking advantage is the broker’s rank correlation minus the principals’ rank correlation (Appendix B §8).

Simulation design and analysis. At the baseline, $N = 1,000$ principals interact for $T = 500$ periods with $\rho = 0.5$, $\delta = 0.5$, and $\eta = 0.02$. One-at-a-time (OAT) sweeps vary the matching problem, turnover, market size, reservation threshold, initial network, broker roster, and self-search pool. Four

two-parameter grids cross ρ with η , ρ with the reservation threshold, η with the reservation threshold, and ρ with δ (Appendix B §9). I refer to each unique model configuration as a regime. Together, the sweeps and grids contain 80 regimes. There are 20 random seeds per regime and 50 at the baseline (1630 runs total).

Neural-network performance depends on the learning rate and the number of training updates, which were calibrated before running the main simulations. I compared candidate settings for principals and the broker separately, considering ranking performance on randomly sampled candidates, stability over time and across random seeds, and computational time.

To show that the findings do not depend on neural-network learning, I repeat the complete design with Ridge regression. In the unrestricted Ridge specification, principals use candidate characteristics, while the broker uses both parties’ characteristics and interactions between them. Ridge penalties were calibrated at the baseline using ranking performance on randomly sampled candidates. Three additional Ridge experiments, conducted at the baseline and across the $\rho \times \delta$ grid, restrict the broker by giving it the same number of observations as a typical principal, retaining only one party’s characteristics from each match, or omitting interactions between the two parties’ characteristics.

To examine the value and structural consequences of assessment and access, I conduct an experiment comparing three versions of the single-broker model. Full service combines the broker’s assessments with access to its contacts and current clients. Assessment-only uses the broker’s assessments to rank candidates available through ordinary self-search. Access-only ranks the broker’s broader candidate set using principals’ own assessments. In each version, the broker still organizes matches, charges the same fee, and observes and learns from the matches it arranges. Each simulation uses one broker model throughout. Changing the service changes matches, outsourcing, network formation, and subsequent learning. The comparisons therefore capture these combined effects.

Ensemble means are averages across random seeds. Window means are taken by first averaging over periods within seeds, and then averaging over seeds. Late-window is $t \in [481, 500]$; early-window is $t \in [51, 70]$. Cross-regime summaries give each regime equal weight. Intervals for differences are calculated within each random seed before being combined across seeds. Unless otherwise noted in a figure caption, time-series figures begin at $t = 30$ to omit the initialization transient. Monte Carlo convergence was assessed in two ways. For each regime and reported outcome, I calculated a two-sided 95% confidence interval for the mean across independent random seeds using the t distribution. I also compared nested estimates based on the first 10 versus all 20 seeds outside the baseline and the first 20 versus all 50 seeds at the baseline. Across the 14 outcomes included in the convergence summary, the median across regimes of the confidence-interval half-width divided by the absolute estimate based on all available seeds ranged from 0.3% to 8.3%. The median across regimes of the absolute difference between the nested estimates, divided by the same full estimate, ranged from 0.1% to 4.8%. These relative quantities are not reported when the full-sample interval includes zero to avoid division by near zero; those cases are assessed in the outcome’s original units.

B The Broker Primarily Provides Assessment, Not Access

Across the tested regimes, the broker is usually the dominant channel in the market. The outsourcing rate as a share of demand is 0.97 at the baseline (late mean, up from an early mean of 0.78) and has a median of 0.96 across regimes. It exceeds one-half in 78 of 80 regimes; the remaining regimes all have no turnover. Yet the broker primarily assesses options within the existing network rather than providing access beyond it. The access fraction (the share of brokered placements between principals with no pre-existing tie) has a late mean of 0.09 across regimes, down from 0.17 early, and never exceeds 0.18. It declines over time in all regimes, and its paired 95% interval is below zero in every regime. Most brokered matches therefore connect principals already tied to each other. Figure 1 and Table 1 summarize these baseline dynamics and the main measures.

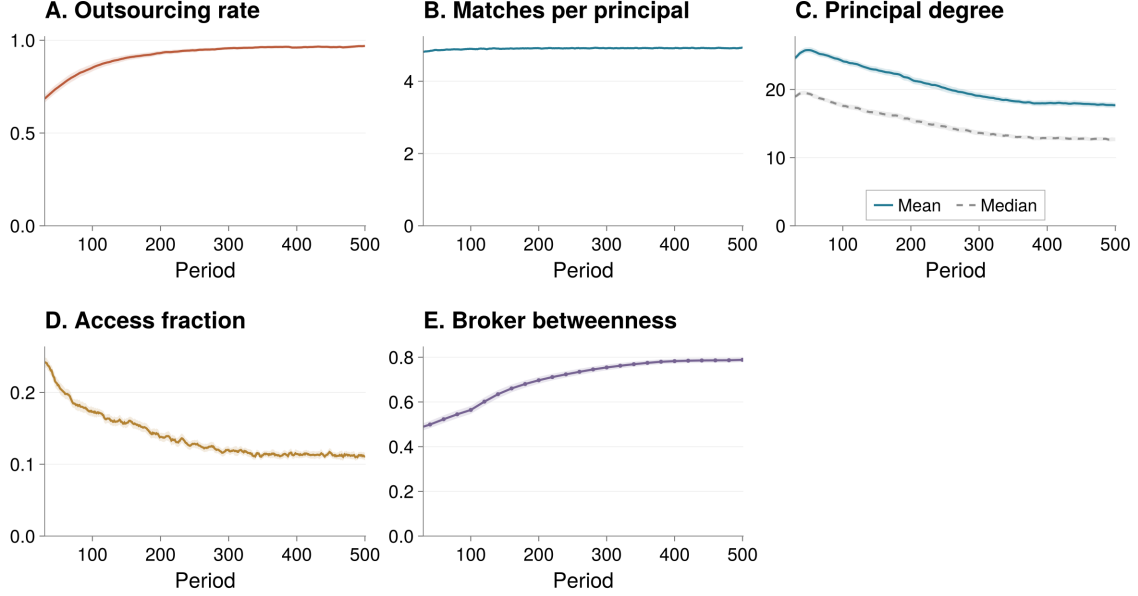


Figure 1: Baseline dynamics. A: outsourcing rate. B: completed matches per principal, counting both participants in each match. C: mean and median principal degree, including links to the broker when present. D: access fraction. E: broker betweenness centrality. Lines are ensemble means after a 5-observation rolling mean within each seed. Ribbons are pointwise 95% Monte Carlo intervals; some are narrower than the lines. Betweenness is measured every 20 periods; other measures are recorded every period.

Table 1: Broker use, access, and assessment quality. Baseline columns report early- and late-window ensemble means; the across-regime column reports the median late-window mean across the 80 regimes. Outsourcing rate is outsourced requested positions divided by all requested positions.

Measure	Baseline		Across regimes
	early	late	late, median
<i>Market use and access</i>			
Outsourcing rate	0.78	0.97	0.96
Output gap (broker – self)	−0.38	+1.71	+1.54
Access fraction	0.20	0.11	0.11
<i>Assessment quality</i>			
Broker rank correlation	0.90	0.95	0.94
Principal rank correlation	0.68	0.63	0.64

At the baseline, the broker’s realized-output advantage develops as the market evolves: the output gap (broker minus self-search) changes from -0.38 early to $+1.71$ late. Across regimes, mean realized output is higher through the broker channel in 66 of 80 regimes, and the late output gap averages $+1.14$. Paired 95% Monte Carlo intervals are above zero in 65 regimes and below zero in 14. The negative gaps occur in 14 regimes, all with no or nearly no turnover ($\eta \in \{0, 0.001\}$). Principals choose a channel using satisfaction scores that include unfilled requests, fees, and search costs, whereas the output gap compares realized output among completed matches. Outsourcing is therefore not itself evidence that the broker always produces better matches.

The broker’s assessment advantage appears more consistently in ranking candidates than in predicting expected output levels. Among randomly sampled potential partners (Appendix A §8), its rank correlation is 0.95 at the baseline and has a median of 0.94 across regimes, compared with 0.63 and 0.64 for principals. The broker-minus-principal difference in rank correlation is positive in 80 of 80 regimes, with paired 95% intervals above zero in 80 regimes. Ranking determines which candidates are considered first, while predicted levels also determine whether offers clear the reservation threshold. Accurate ranking is therefore more important than accurate prediction of expected output levels, although both can affect matching.

Most brokered placements therefore reuse ties that principals could search directly; relatively few expand the principal network. The broker’s ranking advantage appears under pure general quality as well as complementarity: at $\rho = 1$, the broker-minus-principal difference in rank correlation averages 0.26 and is positive in all 8 pure-quality regimes. The OAT checks also show that initial network degree (k_G), broker roster fraction, and the number of strangers available in self-search have comparatively small effects on the main economic outcomes.

C The Value and Structural Consequences of Access and Assessment

The access fraction records how often brokered placements create new ties, but does not establish what makes brokerage valuable to principals. The assessment and access experiment examines these contributions by comparing full service with assessment-only and access-only service.

C.1 Access Can Raise Output While Eroding Structural Advantage

Access can add value even when most brokered placements reuse existing ties. At baseline turnover, net output is 0.242 lower per requested position under assessment-only than under full service (Figure 2, Panel A). This advantage of full service depends on turnover. Without turnover, assessment-only produces more output than full service; at the tested rates of $\eta \in \{0.01, 0.02, 0.03\}$, it produces less. At $\eta = 0.001$, the smallest positive rate tested, the difference is uncertain. The zero-turnover advantage of assessment-only and its losses at $\eta = 0.01$ and $\eta = 0.03$ hold across all tested matching compositions (Supplementary Figure S4, Panel B).

Turnover continually removes principals and relationships, making access to new counterparties useful. Without turnover, relationships accumulate, and candidates reachable through ordinary self-search can support valuable brokerage based on assessment alone. Broader access then provides no net-output advantage. The frequency of introductions alone therefore does not establish the value of access.

Access can also weaken the broker’s structural position. Each successful introduction creates a direct tie that principals can use in later self-search. Providing that connection can therefore reduce the need to buy it again. The comparison with assessment-only supports this self-liquidating mechanism: full service produces higher mean principal degree and lower broker centrality in every tested condition. Principal degree counts each principal’s ties, including its link to the broker when present.

At baseline turnover, mean principal degree initially rises and then declines under full service, whereas it continues to rise under access-only (Figure 2, Panel B). The access-only broker falls further behind full service in centrality over time (Panel D). Its mean centrality difference over the final 20 periods is -0.178 $[-0.196, -0.161]$ (Panel C). Over the same late period, mean principal degree is higher under access-only (25.611 versus 17.727).

At baseline turnover, broker betweenness rises even under access-only service, as continued outsourcing connects principals to the broker. But it remains much lower than under assessment-only service (Panel D). Access therefore exerts a self-liquidating force without producing an absolute decline: introductions create direct ties among principals and erode the centrality the broker would otherwise

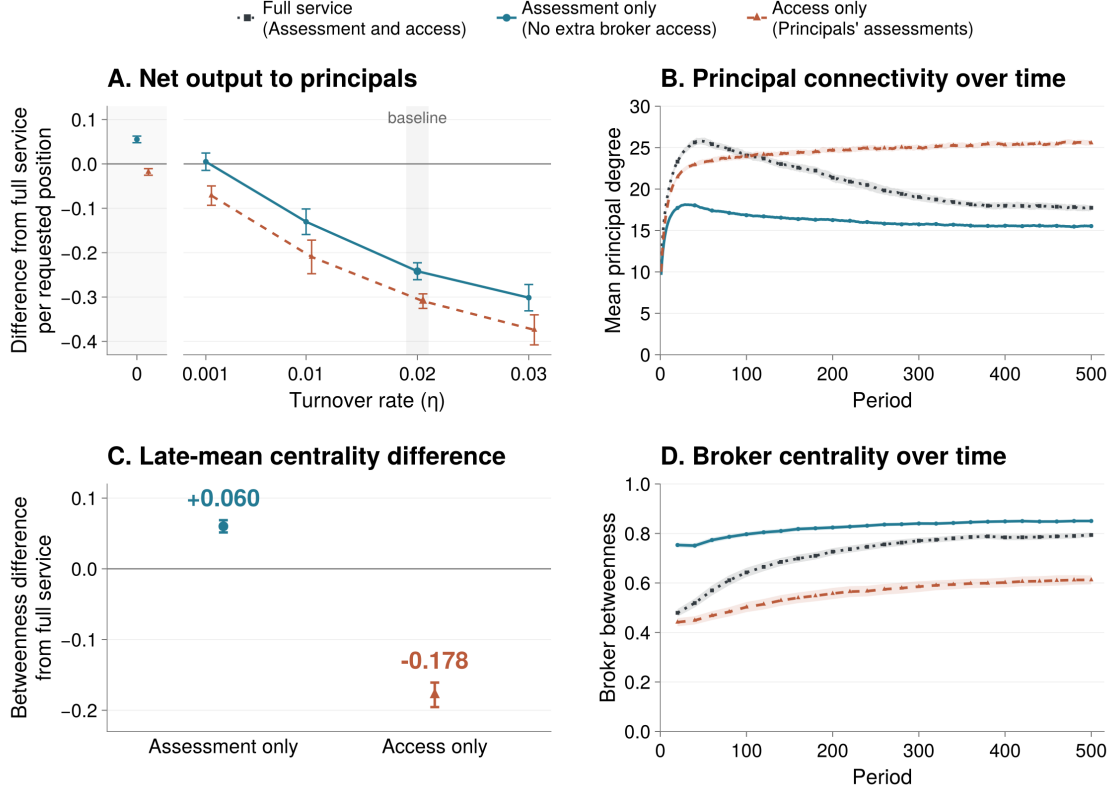


Figure 2: Brokerage with access, assessment, or both. All panels use $\rho = 0.5$ and $\delta = 0.5$. A: assessment-only and access-only minus full service in net output to principals per requested position across 5 turnover rates. Zero turnover is shown separately; positive rates use a linear scale, with baseline turnover shaded. Series are offset horizontally for visibility; lines connect positive-turnover points as visual guides. B–D use baseline turnover of $\eta = 0.02$ per period. B: mean principal degree, recorded each period from $t = 1$ to 500. C: assessment-only and access-only minus full service in late-mean broker betweenness. D: broker betweenness measured at $t = 20, 40, \dots, 500$. B and D join recorded ensemble means without smoothing. A and C show paired differences for $t = 481\text{--}500$; negative values indicate outcomes below full service. Bands and bars are pointwise 95% Monte Carlo intervals. Each model uses 50 seeds at baseline and 20 at other turnover rates; other parameters remain at baseline.

accumulate. Turnover also removes ties and renews opportunities for brokerage. Because learning and outsourcing change with the service provided, these comparisons do not isolate the effect of new direct ties alone.

C.2 Assessment Can Sustain Brokerage Without Additional Access

Principals continue paying for brokerage even when it provides no additional access. At baseline turnover, they outsource 97.1% of requested positions to the assessment-only broker, compared with 96.8% for full service and 82.0% for access-only. The assessment-only and full-service outsourcing rates remain close across the tested turnover conditions, though both fall to about half without turnover.

Adding assessment to access-only service raises outsourcing in every tested condition. Adding access to assessment-only service has smaller effects, sometimes positive and sometimes negative. Assessment is therefore more important than additional access for sustaining brokerage use in this experiment.

Continued outsourcing, however, reflects how brokerage compares with self-search. Principals may keep using the broker because its assessments improve matching, because their independent alternatives are less productive, or both.

The assessment-only broker is already more central than full service at the first measurement. Full service partly catches up over time (Panel D), but the assessment-only broker retains a late-mean centrality advantage of +0.060 [0.051, 0.069] (Panel C). Mean principal degree remains lower than under either other broker model (Panel B), with a late mean of 15.523.

Assessment contributes more to net output for principals than additional access in these comparisons. At baseline, adding assessment to access-only service raises net output per requested position by 0.309, compared with 0.242 from adding access to assessment-only service (Panel A). Assessment-only therefore exceeds access-only by 0.067, with a paired 95% interval of [0.048, 0.086]. This advantage holds in every tested condition, with all paired intervals above zero.

Without the broker’s assessments, access-only produces less net output than full service at every tested turnover rate, including zero. Both assessment and access become more valuable to principals as turnover rises. Between turnover rates of 0.01 and 0.03, output changes little under full service but falls under both assessment-only and access-only.

At baseline, assessment-only has nearly the same outsourcing rate as full service and higher centrality, yet delivers less net output to principals. High centrality need not mean greater use, and neither alone establishes greater value for principals.

The assessment-only result also connects the formal illustration to data enclosure. The broker continues to create value when it assesses only candidates that principals could reach through ordinary self-search. Its information is therefore valuable independently of privileged access. The experiment does not model data enclosure itself, because the broker still mediates matches and learns from them. It nevertheless establishes its prerequisite: assessment can become a distinct service that could be unbundled from intermediation and sold as predictions or analytics.

D The Broker’s Information Advantage Comes from Data Volume and Relational Information

The theory identifies two sources of the broker’s information advantage: the greater volume of observations generated by working across principals and relational information about both parties to each match. The Ridge experiments separate these sources from the flexibility of the neural network. When Ridge regression replaces both neural networks, the broker-minus-principal difference in rank correlation is 0.374 at the baseline, has a median of 0.354 across the 80 regimes, and is positive in all 80. The full Ridge model still gives the broker more observations and more information about each pair than a principal. Because each restriction changes later matching and learning, their effects need not be additive.

Data volume accounts for much of the advantage (Fig. 3). Across the 31 regimes in the baseline and $\rho \times \delta$ design, the median broker-minus-principal difference in rank correlation is 0.440 in the full Ridge model but only 0.056 when the broker trains on the same number of observations as a typical principal. The difference remains positive in 27 of 31 regimes under this restriction, so data volume is important without being the broker’s only advantage.

The content of the broker’s data also matters. When each observation retains the characteristics of only one party to a match, the broker’s median advantage over principals in ranking new candidates is

-0.253 and remains positive in only 8 of the 31 tested regimes. Retaining both parties’ characteristics improves on this one-party version in 30 of 31 regimes, and retaining how those characteristics interact improves it further in 30 of 31 regimes. Interaction information matters most when complementarity contributes strongly to expected match output and not at all under pure general quality. The broker’s advantage therefore comes from both the volume of observations generated through repeated vicarious relational work and information about how the characteristics of two parties combine.

In the assessment and access experiment, the access-only broker also develops an informational advantage. Although it uses principals’ assessments to choose matches, it continues to record and learn from their outcomes. When tested on randomly sampled potential partners, its own predictions rank potential partners by expected match output more accurately than principals’ predictions (baseline rank correlations of 0.982 and 0.708, respectively). Under the model’s observation and learning rules, the broker thus gains an informational advantage even though it does not use its own assessments in matching. The assessment-only result shows that this capability can sustain a paid service without additional access.

E The Nature of the Matching Problem Shapes the Broker’s Advantage

The full-service comparisons vary two features of the matching problem: whether expected match output depends on general quality or pair-specific complementarity, and how difficult complementarity is to learn. The parameter ρ moves expected match output from pure complementarity at $\rho = 0$ toward pure general quality at $\rho = 1$. The parameter δ increases difficulty by making expected output change more sharply across a boundary in type-pair space. Because complementarity disappears at $\rho = 1$, δ has no effect there. The pure-quality case is therefore a separate boundary condition, not a smooth continuation of the interior ρ values.

Broker centrality is governed primarily by the composition of expected match output. In the one-at-a-time sweep from $\rho = 0$ to $\rho = 0.85$, the access fraction rises from 0.10 to 0.14 and broker betweenness

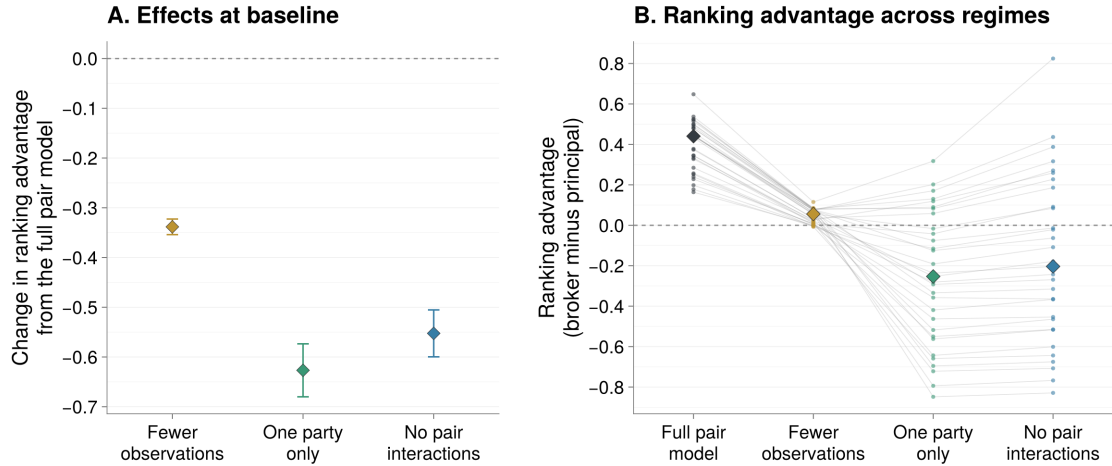


Figure 3: Sources of the broker’s ranking advantage. Ridge broker models described in Model and Methods. A: each restriction minus the full pair model in late-window ranking advantage at baseline. Diamonds are mean differences paired by random seed; whiskers are 95% Monte Carlo intervals. Negative values indicate a smaller ranking advantage. B: late-window ranking advantages across the 31 regimes in the $\rho \times \delta$ design. Small points are regime means, lines connect the same regime across models, and diamonds mark medians across regimes. Dashed lines mark zero.

centrality falls from 0.86 to 0.51. The $\rho \times \eta$ grid shows the same directions at every turnover rate. At the pure-quality boundary, centrality falls further to 0.15, while access returns to 0.11. The boundary therefore extends the decline in centrality but not the increase in access below $\rho = 1$.

Under pure complementarity, greater difficulty lowers the ranking accuracy of both the broker and principals, but it does not eliminate the broker’s advantage. Increasing δ from 0 to 1 lowers the broker’s rank correlation from 0.92 to 0.69 and the principal correlation from 0.71 to 0.46 (Fig. 4). The corresponding broker-minus-principal differences are 0.21 and 0.23. The size of this difference is not monotonic in difficulty, but its paired 95% interval is above zero in 31 of the 31 regimes. Because δ changes only complementarity, it is inactive at the pure-quality boundary.

The broker’s realized-output advantage is greatest when expected match output depends more heavily on complementarity. Paired 95% intervals for the broker-minus-self-search output gap are above zero in 31 of the 31 regimes. Composition and difficulty therefore shape different outcomes. Greater difficulty makes candidates harder for both sides to rank but does not produce a monotonic change in the broker’s relative ranking advantage. Greater reliance on complementarity increases the mean realized output produced through brokered matches. The broker retains both a ranking advantage and an output advantage throughout the grid.

The assessment and access experiment provides direct evidence for this role of complementarity. At each turnover rate compared across matching compositions, adding the broker’s assessments to access-only service produces a larger gain in net output under pure complementarity than under pure general quality (Supplementary Figure S4). Complementarity therefore increases the value to principals of using the broker’s assessments, beyond the access that the broker already provides.

F A Broker Can Be Highly Central While Doing Little Bridging

In most full-service regimes, the broker’s structural position strengthens over time while principals have fewer direct ties on average. At the baseline, mean principal degree falls from 25.4 to 17.7, while broker betweenness centrality rises from 0.55 to 0.79 (early to late means; Fig. 1). This combination occurs in 64 of 80 regimes, and principal degree falls in 64 of them. In most regimes, direct ties do not accumulate fast enough to offset turnover. The broker’s structural advantage therefore does not disappear as brokered matches create direct ties. Because outsourcing connects current clients to the broker, sustained use converts the value of assessment into structural centrality.

The distinction between assessment and access helps explain this relationship. An access match creates a direct tie between previously unconnected principals and can remove the broker from their shortest path. An assessment match reuses an existing tie, so the broker can serve the market without directly adding ties to the principal network.

The assessment and access experiment provides direct evidence of these structural consequences. At baseline turnover, full service produces higher mean principal degree and lower broker centrality than assessment-only (Figure 2, Panels B–D). Providing access alongside assessment therefore strengthens principals’ connectivity while weakening the broker’s structural position.

This helps explain why the full-service broker usually gains centrality in the main simulations. As its work becomes increasingly assessment-based, relatively few brokered matches create new ties among principals. Valuable assessments sustain outsourcing, which connects principals to the broker, while turnover removes ties and renews opportunities for brokerage. The broker’s structural position strengthens despite the self-liquidating force of its introductions.

Figure 5 compares broker centrality with principal degree (Panel A) and access fraction (Panel B) across matching compositions and turnover rates. Holding turnover fixed, moving from pure complementarity toward $\rho = 0.85$ raises access and lowers centrality. Holding ρ fixed, increasing

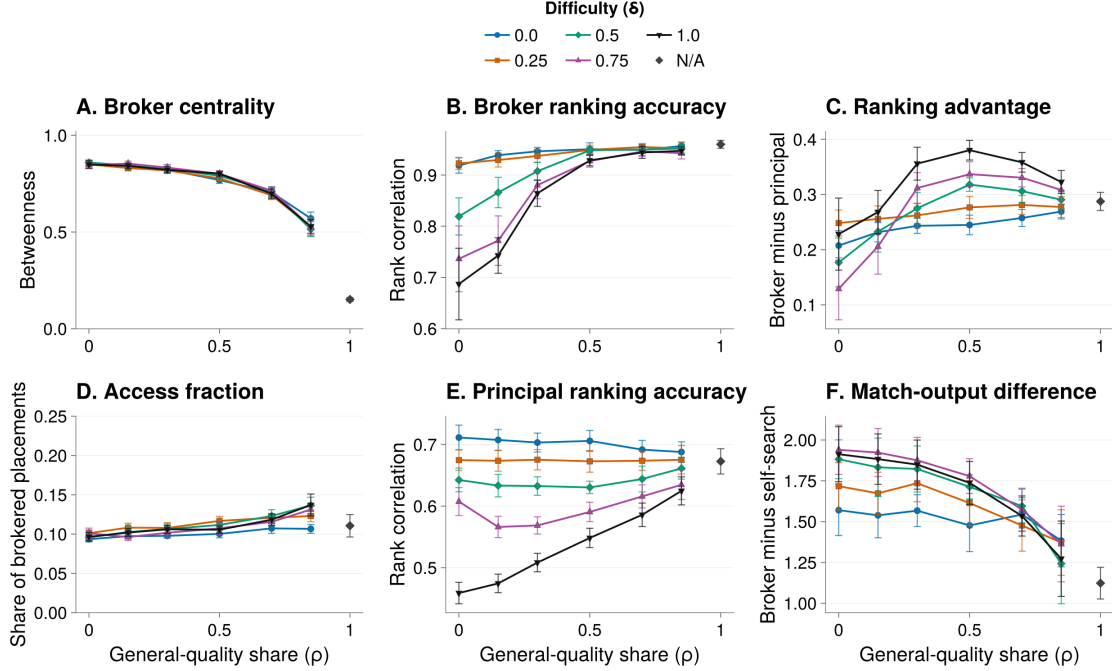


Figure 4: Outcomes across the matching problem ($\rho \times \delta$). Late-window means across all 31 regimes in the $\rho \times \delta$ design, with other parameters at baseline. A: broker betweenness. B and E: broker and principal rank correlation. C: ranking advantage, broker minus principal. D: access fraction. F: mean realized output per completed match, broker minus self-search, before costs and fees. Colors and symbols identify difficulty δ . N/A denotes the unconnected gray diamond at $\rho = 1$: difficulty affects only complementarity and is not applicable under pure general quality. Whiskers are 95% Monte Carlo intervals.

turnover raises both access and centrality at every tested ρ . Turnover creates network gaps, while matching composition determines how often brokerage closes them with direct ties.

The turnover effect is substantial but nonlinear (Panel A). Holding other parameters at baseline, increasing η from 0 to 0.03 lowers late mean principal degree from 77.4 to 16.3 and raises broker centrality from 0.13 to 0.82, with most of the change occurring between $\eta = 0.001$ and 0.01. Turnover also changes the broker's realized-output advantage. In the $\rho \times \eta$ grid, the 95% interval for the broker-minus-self-search output gap is below zero in 5 of 5 ρ conditions at $\eta = 0$, but above zero in 5 of 5 by $\eta = 0.01$. In the model, turnover removes principals and ties, creating network gaps and the conditions under which brokered matches outperform self-search.

The pure-quality boundary confirms that low access alone is not sufficient for high centrality. Although higher turnover raises centrality there, centrality remains below pure complementarity at the same turnover rate. In the tested regimes, high centrality occurs when turnover renews network gaps and brokerage mainly assesses existing relationships.

Figure 6 shows that structural position relates differently to relative ranking and realized match output. Holding all other parameters fixed, moving from pure complementarity to pure general quality reduces the output gap from +1.27 to +0.54, while increasing the broker-minus-principal difference in rank correlation from 0.18 to 0.26. The broker is therefore most central and produces its largest realized-output advantage where its advantage over principals in ranking candidates is smaller.

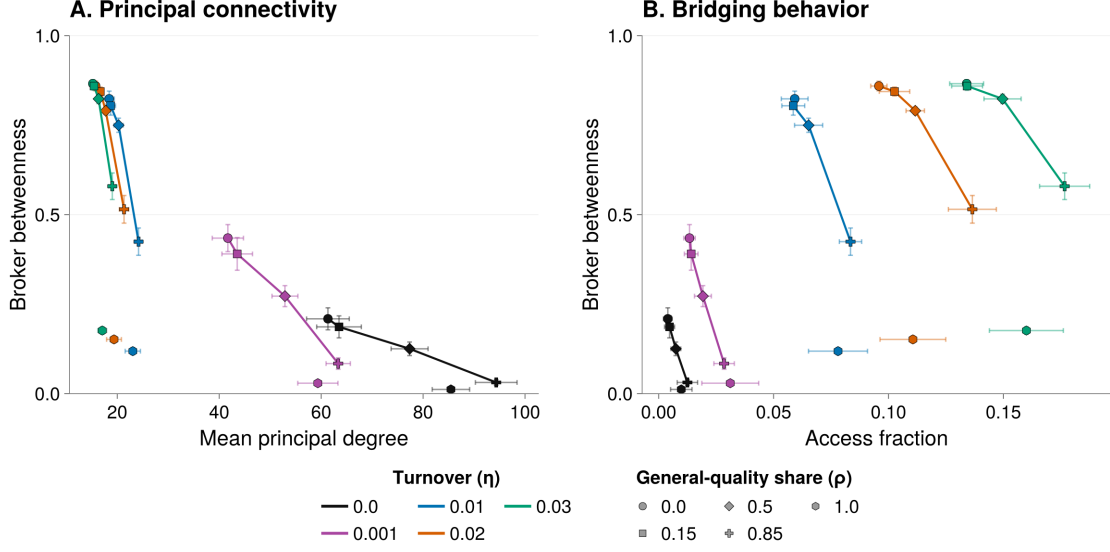


Figure 5: Broker centrality, principal connectivity, and bridging behavior. A: broker betweenness against mean principal degree, including links to the broker when present. B: broker betweenness against access fraction. Points are late-window means for the same 25 combinations of general-quality share ρ and turnover η , with other parameters at baseline. Color identifies η , shape identifies ρ , and whiskers show pointwise 95% Monte Carlo intervals on both axes. Lines connect $\rho < 1$ within each turnover rate as visual guides; the pure-general-quality boundary ($\rho = 1$) is unconnected. Panels share the same vertical scale.

Repeated relational work generates information that can sustain the broker's assessment advantage. When those assessments remain valuable relative to self-search, principals continue outsourcing even when they can reach counterparties themselves. Sustained use converts the value of assessment into structural centrality. High centrality can therefore be a product of the broker's relational work rather than evidence of frequent bridging.

G Propositions

The simulation results motivate the following falsifiable propositions for future empirical research:

P1. *Assessment is the primary source of brokerage value.* In matching markets, brokers add value primarily by assessing counterparties rather than by bridging disconnected parties. The value of this assessment role increases as complementarity contributes more to match output.

P2. *Relationship disruption increases the value of access brokerage.* Clients gain more from brokers' ability to find new counterparties when market participants frequently leave and are replaced than when they can rely on stable relationships.

P3. *When the broker records and accumulates data about his matching work, brokerage widens informational asymmetry.* Repeated brokerage increases the broker's assessment advantage over his clients, even though those clients also learn from the matches he arranges.

P4. *Assessment brokerage produces high broker centrality with little bridging behavior.* Sustained outsourcing converts the value of assessment into a greater structural centrality for the broker. Brokers whose assessments remain valuable relative to clients' independent alternatives become highly central even when most of their matches involve counterparties already connected to one another.

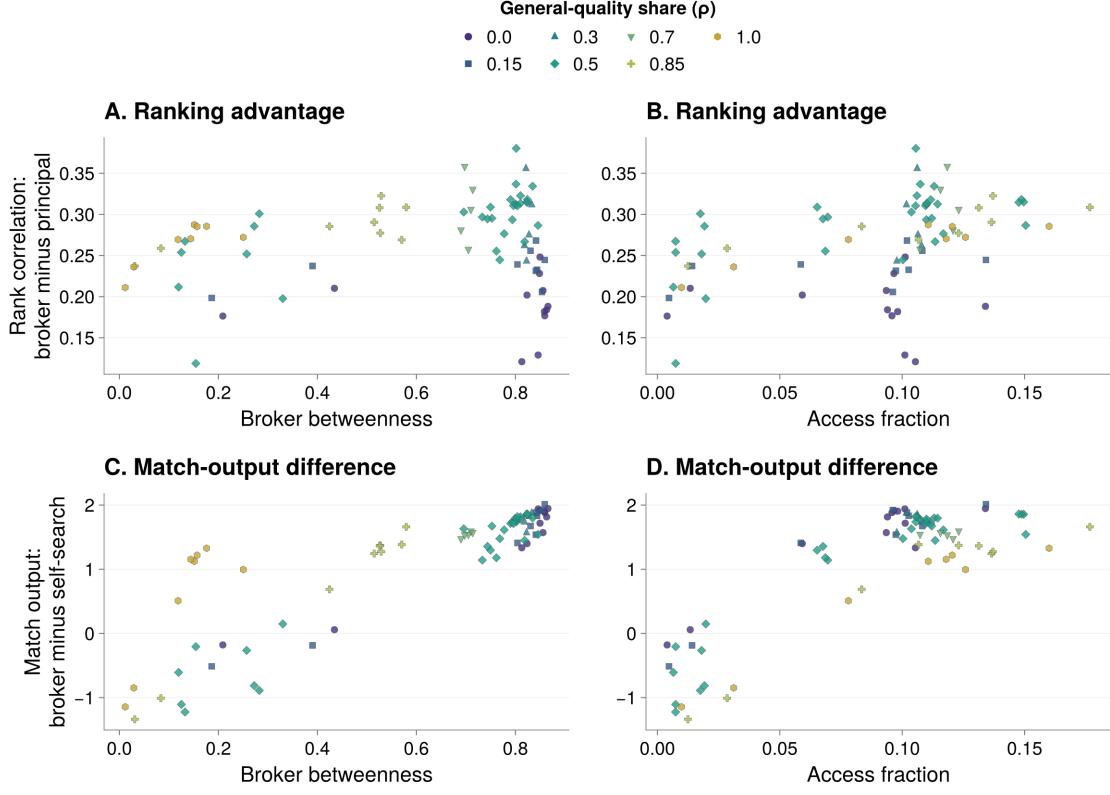


Figure 6: Ranking and output differences by centrality and access. A–B: ranking advantage, broker minus principal. C–D: mean realized output per completed match, broker minus self-search, before costs and fees. Horizontal axes show broker betweenness (A, C) or access fraction (B, D). Each point is a late-window mean for one of the 80 tested regimes. Colors and symbols identify general-quality share ρ .

P5. *Assessment enables the broker to become a principal.* Corporate brokers whose assessments create value independently of privileged access are more likely to use that capability on their own behalf, either to acquire and sell resources they previously intermediated (resource capture) or to sell predictions or analytics directly (data enclosure).

V Transient Brokerage and Its Conditions of Possibility

The model formalizes how brokers accumulate and sustain informational and structural advantages, but it stops before the transition to principals. I now return to the theoretical argument to explain how a corporate broker may convert those accumulated advantages into resource capture or data enclosure.

Outsourced relational work performed at scale by corporate brokers creates the conditions under which this conversion can occur. The broker dismantles the relational package previously negotiated with its clients and replaces it with a new configuration organized around the broker’s ownership of the resources it used to intermediate (*resource capture*), or the monetization of the information generated through prior matching activity (*data enclosure*).

These configurations are two qualitatively different forms of capture, which can be pursued in parallel and along with standard brokering. In *resource capture*, the broker takes inventory risk and starts acting as a principal in the market it used to intermediate. Counterparties interact only with the broker, without creating direct ties with one another. As the broker expands its principal activity and shrinks its matching services, former clients become counterparties or competitors. In *data enclosure*, the broker monetizes information directly by selling predictions, analytics, or other information services, often as a subscription.

The assessment-only experiment provides a formal precursor to this transition. In that experiment, the broker no longer provides privileged access, yet its assessments continue to improve matching and sustain outsourcing. Data enclosure takes this separation one step further: the broker monetizes its accumulated assessment capability without necessarily continuing to mediate the transactions it evaluates.

What makes capture possible, I suggest, is primarily the broker’s information advantage, not its structural position. Structural and informational advantages are decoupled, and capture may occur even if the broker’s centrality is declining. This is because constructing viable relational packages generates an informational byproduct that the broker can leverage. Two features of the broker’s informational landscape facilitate capture: cumulative knowledge of complementarities coming from the *outsourced* nature of its relational work, and information infrastructures arising from its *corporate* nature. Other actors cannot easily replicate either advantage.

(i) *The role of complementarity information.* The broker’s vicarious work generates knowledge of how different parties pair together, what kinds of matches succeed across the market, and what relational packages support different transactions. For regular market actors, this complementarity information is hard to replicate. Unlike attributional information about the general quality of counterparties, which any actor with sufficient market exposure can in principle acquire, complementarity information is generated through the repeated observation of various pairings. The broker holds it because it does the matching for a variety of clients; clients, who experience only their own matches, do not—unless they purchase or otherwise acquire it. Complementarity information supports resource capture, because it allows the broker to identify which resources or counterparties are most likely to generate profitable matches, so that it can capture them strategically and offload them gainfully. Complementarity information also supports data enclosure because it can be packaged into prediction or analytics products and services.

(ii) *The role of information infrastructures.* The corporate broker’s information infrastructure explains *how* its information advantage can be preserved and converted. The taxonomies, classification systems, contractual templates, and algorithmic sorting procedures that the broker develops to perform vicarious work at scale and in a distributed manner, through the everyday work of its employees and representatives, are also the infrastructures through which capture can be realized. They can be repurposed to route resources to the broker itself, to put these resources back in circulation profitably, or to organize data into a marketable product. Moreover, the corporate form gives the broker the legal privileges required to build, own, and protect these infrastructures over time (Coleman 1982; Pistor 2019).

These features, and transient brokerage itself, can be found across markets. I illustrate each configuration with an example.

Dealers in collectible markets. Collectors of art, fine wine, rare books, or similar specialty goods may seek trades through broker-dealers who know the market. Each collector has distinct tastes and holdings; match quality depends on multidimensional complementarity between what one party has and what another wants. Dealers accumulate knowledge of collector preferences across transactions, and they may decide to leverage it and start holding inventory, for instance, by launching and operating a gallery.

Interdealer brokerage in over-the-counter (OTC) financial markets. Interdealer brokers (IDBs) are specialized intermediaries in OTC financial markets (e.g., bond markets, foreign exchange, commodities). IDBs help institutions like banks and dealers conclude large, complicated, or anonymous transactions in the absence of a centralized trading place. They solve a hard matching problem: the right counterparty for a trade depends on inventory positions and risk appetites that parties wish to keep secret. IDBs must know who wants what, negotiate prices across hidden order interest, preserve anonymity until execution, and manage long-term client relationships to remain in the order flow. The complementarity information they accumulate, about who will absorb what positions at what price under what conditions, is substantial. Resource capture, however, faces regulatory and reputational hurdles. Instead, IDBs have monetized accumulated information through data divisions that package historical OTC pricing and other data products and sell them to the same dealers whose trades generated the data and to a wider set of buy-side firms and regulators (see, e.g. Parameta Solutions 2026; TP ICAP 2026).

In summary, three elements enable capture or enclosure: a matching problem that makes complementarity information valuable, an informational advantage accumulated through repeated vicarious relational work, and an institutional architecture that allows that information to be aggregated, organized, and protected.

VI Discussion

A Structural Advantage as a Product of Relational Work

Within the agent-based model, the broker’s centrality is a product of its relational work. The broker is central because many principals outsource to it repeatedly: good assessments help keep brokerage attractive relative to self-search. Its centrality rises and falls with outsourcing and satisfaction, not with exogenous features of the initial graph.

The simulations show that the broker’s relational work endogenously produces its structural advantage. This process generates two patterns that existing theory does not readily predict. The first is that the broker tends to be most central exactly where it bridges least in practice. Across variation in matching composition at a fixed turnover rate, centrality is negatively associated with the access fraction (Figure 5B): in the complementarity-dominated markets where the broker is most prominent, almost all its matches re-pair principals already tied to one another. A strong bridging position coincides with weak bridging behavior. Occupying a strategic position and using it for one’s work are different things that theory should not conflate.

The second pattern concerns the sustainability of brokerage. A long-standing expectation is that brokerage tends to undermine its own structural basis (Burt 1992, 2002; Stovel, Golub, and Milgrom 2011). Each successful introduction creates a direct tie that can reduce future dependence on the broker. Access therefore creates a self-liquidating force, not an inevitable decline in broker centrality.

The experiment makes this distinction clear. At baseline turnover, full service is less central than assessment-only, yet all three broker models gain centrality over the observed period (Figure 2, Panels C–D). Providing access can thus lower centrality relative to assessment-only without causing centrality to decline over time. Turnover removes ties and renews opportunities for brokerage, while valuable assessments can sustain outsourcing. A broker can therefore become more central even while its introductions weaken one source of its structural advantage.

The results also suggest a precise mechanism for the decoupling of structural and informational advantage posited in Section V. The two advantages respond to different features of the matching problem. The broker’s structural prominence tracks the composition of match value, rising as matching becomes more complementarity-driven, while its prediction edge responds to the difficulty

of learning the complementarity component, which can vary independently. A broker can be highly central despite a modest informational lead or hold a decisive informational lead from an unremarkable position. Betweenness centrality is best understood not as a static measure of opportunity but as a product of the relational work the broker performs.

B Scope Conditions and Generalizability

The concept of transient brokerage applies most directly to matching markets in which (i) matching is complex enough that the value of a match depends not only on the general quality of each party but also on their complementarity and (ii) a corporate broker operates as a repeat-player accumulating information across transactions. The further a case departs from these conditions, the weaker the predictions of the theory. When the relational dimension is weak, the broker’s advantage rests primarily on data volume about multiple counterparties, making its informational position less privileged. When the broker is a physical person, she lacks the temporal persistence, infrastructural capacity, and legal architecture that the corporate broker relies on.

The dynamics of transient brokerage also depend on the nature of the brokered resource and the institutional environment. In a goods market where items are bulky and physical delivery is required, the costs and logistics of capture differ substantially from those in a market for services or a market where goods are virtual or ownership passes without physical possession. In some markets, regulation may slow or prevent capture. Market structure matters as well. The model considered in this article features a single broker. With multiple brokers competing for the same clients or matches, the cross-market data pool is fragmented, and each broker may accumulate a smaller informational advantage, unless one or a few brokers dominate the market. A systematic treatment of broker competition is deferred to future work.

Transient brokerage is relevant to a particular configuration of brokerage, in which the matching problem, corporate form, and information accumulation give rise to a distinctive trajectory. The sociological literature has used brokerage to describe a wide range of phenomena, from immigrant children translating for their parents to marriage matchmakers and political brokers. The relational-work view of brokerage developed in this article can be applied to what many of these so-called brokers do. Without the repeat-player structure of corporate brokerage, the legal and infrastructural architecture of the firm, and an economic matching problem complex enough to sustain an informational advantage, however, the pathway to capture is not the same.

These scope conditions point to candidate domains in which to test the theoretical propositions. Real estate and various types of financial intermediation, including interdealer brokerage, combine complex matching, corporate repeat-player brokers, and meaningful complementarity information. Import-export management and data markets also share many of these features.

C The Political Economy of Corporate Intermediation

The corporate broker is a vehicle for the accumulation of information generated by the interactions between others, and the principals whose matching activity produced the information do not participate in the profits. The asymmetries that make capture possible are not just a byproduct of a broker’s size or expertise. They are legally constituted. Corporate brokers assemble cross-market data and build classification systems and information infrastructures under the legal protections of the corporate form, which allows vicarious relational work to solidify into a durable informational moat. Whether a broker can convert its informational position into resource ownership or data provision depends on the institutional environment as much as on network position.

When the broker becomes a principal, it changes who gets connected to whom. The principals observe far fewer matches than they otherwise would, and most matches in the market teach them nothing

at all. The informational asymmetry that makes capture possible is widened by capture itself: the broker takes control not only of the resource but of the data its placement generates. Transient brokerage shifts how value is distributed, channeling the returns of participants’ collective relational activity to the corporations that record and mine the data it generates.

Capture by corporate brokers has become more common since the 1980s alongside three transformations that have reshaped market intermediation: the turn to finance, which expanded the share of economic activity routed through specialized intermediaries (Carruthers 2015; Krippner 2011); the digital turn, which dramatically lowered the cost of accumulating, storing, and analyzing the data that brokerage generates while also expanding the demand for predictive analytics (Cohen 2019; Gandy 1993); and the rise of platform capitalism, which produced multi-sided market operators that own the relational infrastructure of matching while monetizing the data it generates (Sadowski 2020; Viljoen, Goldenfein, and McGuigan 2021). Transient brokerage is one of the mechanisms that link these transformations together and contributes to a broader shift in how value is extracted from economic activity.

Significant evidence from financial markets shows that corporate brokers that once facilitated matches between others have increasingly become principals, either by capturing the activity they used to intermediate or by monetizing data. The demutualization of stock exchanges is a prime example. As late as the mid-1990s, almost all major global exchanges operated as member-owned mutual organizations. By the mid-2000s, a majority had introduced electronic trading systems that allowed them to grant trading access directly to investors, bypassing the services of member firms, and had converted to for-profit shareholder structures (Aggarwal 2002; Petry 2021). For-profit exchanges now derive a growing share of revenue from proprietary data feeds and trading technology (MacKenzie 2021; Pardo-Guerra 2019).

A parallel transformation has taken place in interdealer brokerage for over-the-counter (OTC) markets, where the data divisions of prominent brokers now generate substantially higher operating margins than the broking activities themselves; this is the case, for instance, of the Parameta Solutions data division of TP ICAP, the largest interdealer broker (TP ICAP 2026). In consumer markets, Visa offers another example. Reorganized in 1970 as a member-owned cooperative of issuing banks, Visa operated for decades as a non-profit utility that brokered payment authorizations between cardholders, merchants, and banks (Stearns 2011). Following its 2007 restructuring and 2008 IPO, Visa converted into a for-profit corporation that now derives substantial revenue from data, analytics, and risk solutions services, alongside the network and processing fees generated by its core authorization and clearing activities (Visa Inc. 2024).

VII Conclusion

This article reconceptualizes brokerage as outsourced relational work and explains how repeated matching can produce an informational advantage that a corporate broker may later convert into resource capture or data enclosure. The formal illustration demonstrates one possible pathway through which repeated vicarious relational work can produce informational and structural advantages, and it generates propositions about when those advantages should arise. By pooling observations across clients, the broker learns how both parties’ characteristics combine and develops assessments that are particularly valuable when match output depends on complementarity. Those assessments can sustain outsourcing even when principals can reach counterparties themselves. Continued outsourcing connects principals to the broker and can increase its structural prominence. When the broker arranges matches between previously unconnected principals, it also creates direct ties that can weaken that position. The broker can therefore be highly central while doing little bridging.

The very conditions that make brokerage possible and valuable, such as the scarcity of information and the complexity of the matching problem, create opportunities for the broker—particularly the corporate broker—to take control of resources. At the transition point, the broker dismantles the relational package negotiated with its principals and collapses the boundary between *vicarious* and *personal* relational work to pursue its own advantage. It abandons or reconfigures the service-tie configuration that existed and (re)negotiates a new one, premised on the broker’s direct ownership and control of formerly brokered resources or the data they generated. In the latter case, the accumulated data can also be mined internally or licensed out to feed proprietary behavioral prediction, classification, and scoring products (Fourcade and Healy 2017, 2024) for use in markets distinct from those in which they originated (Viljoen 2021; Zuboff 2019).

This article helps explain why the digital turn has facilitated broker capture at scale. Because information is central to brokerage, changes in the information ecosystem reshape the conditions under which transient brokerage becomes available, and digital technology has changed those conditions decisively. Relational infrastructures that once consisted of internal handbooks, contractual templates, and bureaucratic procedures now incorporate data taxonomies, algorithmic sorting, and interface designs that constitute the relational packages they administer in real time and at scale. Digital technology amplifies these asymmetries and shortens the path from accumulated information to ownership or monetization.

Digital platforms and algorithmic marketplaces partly automate the work of matching through auction protocols, ranking algorithms, and rating systems. Where platform operators perform this work on behalf of others, automation reorganizes vicarious relational work rather than eliminating it. The operator administers categories, ratings, and defaults that shape how users interact and transact. It also accumulates information across users. It can leverage this information to acquire and sell resources on its own account (resource capture), or to sell predictions and analytics directly (data enclosure). Both activities can coexist with continued brokering. Brokerage becomes transient when matching on behalf of others becomes secondary or ceases.

For example, Amazon operates as both a marketplace for third-party sellers and a direct retailer of its own products, showing how these roles can coexist. Zhu and Liu (2018) found that Amazon was significantly more likely to begin directly selling products with higher sales and better customer reviews, using information available to it as the marketplace operator. This illustrates how information accumulated through intermediation can guide the operator’s own transactions, including competition with the sellers it serves. Brokering continues alongside direct selling. The broker can strengthen its advantage while becoming a competitor to the clients whose transactions helped produce it.

Two main directions warrant future work. On the modeling side, the ABM developed here is exploratory. It has many interacting components and considerable researcher degrees of freedom in its specification, and it carries the usual caveats of this kind of hand-crafted simulation: its results illustrate the theory rather than establish it. Systematic investigation of the parameter space, the decision rules, and the model’s sensitivity to them may surface dynamics that the results presented in this article do not reveal. The resource capture and data enclosure mechanisms are introduced theoretically but not formalized; they are natural extensions. On the theoretical side, the framework does not yet fully address how platforms and algorithmic marketplaces reshape the dynamics of brokerage by automating the vicarious relational work that brokers historically performed. Both directions point toward the same underlying question about the impact of the digital turn on brokerage, the institutions that rely on it, and the actors that perform it or benefit from it.

The digital turn has brought about a series of transformations. The informational byproduct of relational matching has come to rival the value of the matches themselves. Relational work is increasingly valued for the data it generates and performed by machines rather than people. The informational asymmetry between brokers and their principals has widened. Existing relational

packages are being dismantled to extract value from the information and infrastructures associated with them. These shifts redraw the boundaries on which brokerage rests: between internalized and outsourced relational work, and between vicarious and personal relational work. Brokerage becomes a transient position, held so long as brokering remains more profitable than capture.

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